



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -1 (Paper-1)

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Programming Concepts

❖ Computer Program

- Computer programs are the collections of instructions that tell a computer **what to do**, how to interact with the user, and interact with the computer hardware and process data.
- Computer programs, known as software.
- Without programs, a computer is an empty machine.
- Computer do not understand human languages, so you need to use computer programs to communicate with them.

❖ Programming

- The Process of written program.

❖ Programmer

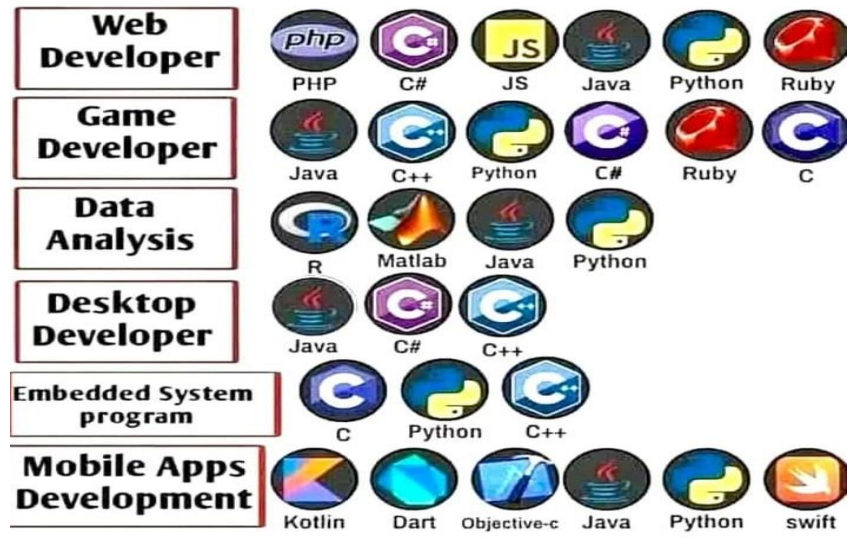
- A programmer is a person who writes computer programs using programming language to solve problems or perform tasks.

❖ Programming Languages

- Programs are written using programming languages.
- Programming can be done using a variety of computer "languages,"
E.g.
C&C++,Java,Python,COBOL,GO,JavaScript,R,SQL,PHP,Ruby,Fortran,VB,...



Programming Languages You Should Learn To Become



❖ Syntax error

- Syntax refers to the spelling and grammar of a programming language.
- Program with syntax errors cannot execute.

❖ Logic & logical error

- A logic error (or logical error) is a mistake in a program's source code that results in incorrect or unexpected behavior.
- It is a type of runtime error that may simply produce the wrong output or may cause a program to crash while running.

Syntax Error	Program Not run
Logical Error	Program run, but output wrong/incorrect/unexpected

❖ Programming Steps

The following steps in the Program Development Life Cycle:

1. Analyze the problem

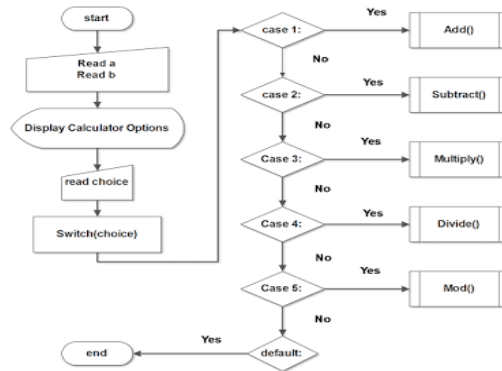
The computer user must figure out the problem, then decide how to resolve the problem - choose a program.

2. Algorithm Design

In this stage all the instructions which are to be performed at different stages in the program are listed in simple English language.

3. Flow Chart / Design the program

A flow chart is important to use during this step of the PDLC (Program development life cycle). This is a visual diagram of the flow containing the program. This step will help you break down the problem



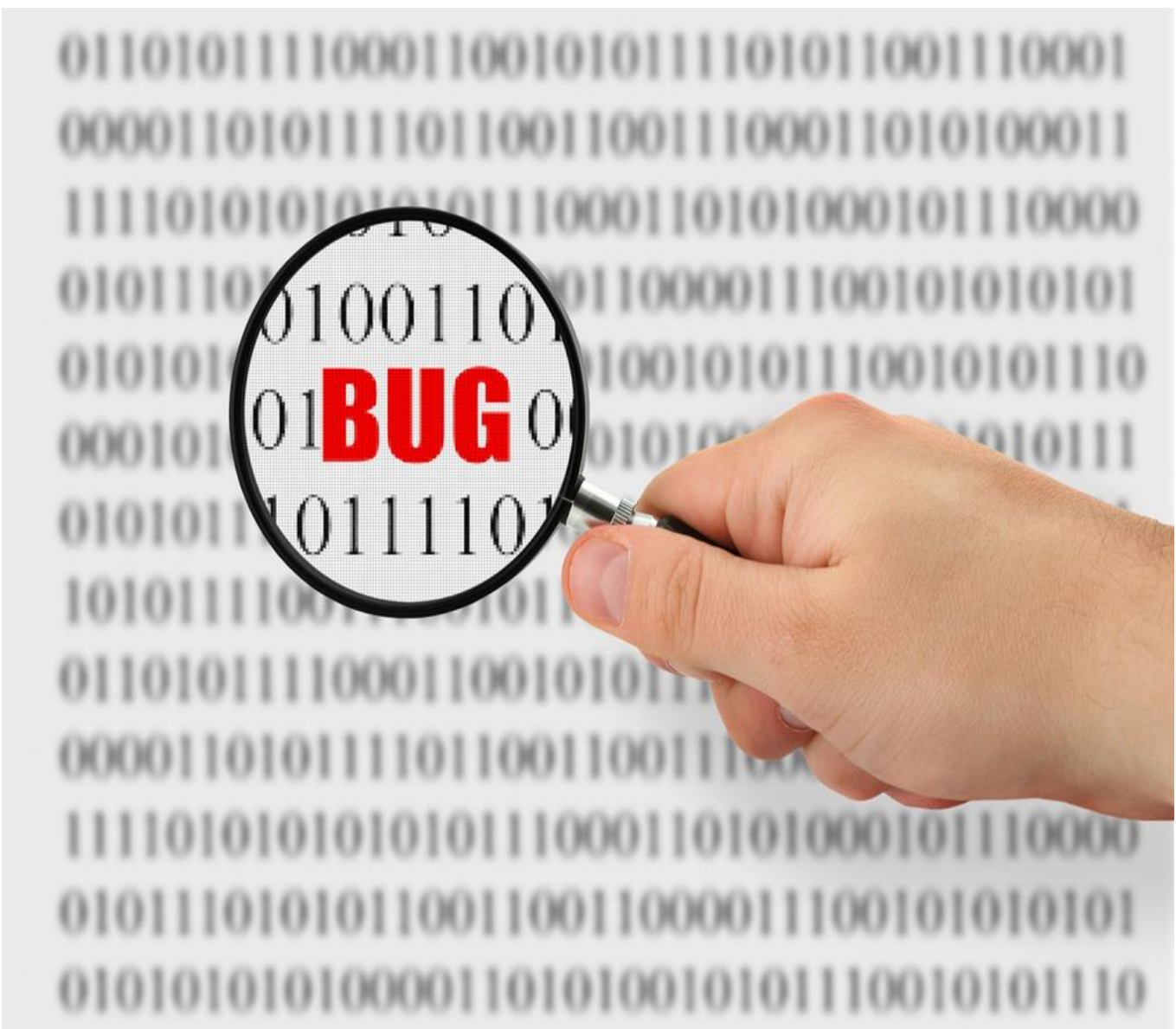
4. Coding / Code the program

This is using the language of programming to write the lines of code. The code is called the listing or the source code. The computer user will run an object code for this step.



5. Debug the program

The computer user must debug. This is the process of finding the "bugs" on the computer. The bugs are important to find because this is known as errors in a program.



6. Formalize the solution / Testing

One must run the program to make sure there are no syntax and logic errors. Syntax are grammatical errors and logic errors are incorrect results.

7. Document and maintain the program

This step is the final step of gathering everything together. Internal documentation is involved in this step because it explains the reasoning one might have made a change in the program or how to write a program.

❖ Characteristics of a Good Program

- **Efficiency:** A good program should use minimal resources to perform its tasks.

- **Reliability:** It should consistently produce accurate results under various conditions.
- **Maintainability:** The program should be easy to understand, update, and modify.
- **Scalability:** It should be able to handle increased workloads without major changes.
- **Security:** A good program should protect data and prevent unauthorized access.
- **Usability:** The program should be intuitive(अन्तर्यामी) and user-friendly.
- **Portability:** It should be able to run on different platforms without major modifications.
- **Readability:** The code should be clear and well-organized for easy comprehension.
- **Modularity:** The program should be divided into manageable and reusable components.
- **Documentation:** Good programs have comprehensive and clear documentation for users and developers.

❖ What is an algorithm?

- The step-by-step procedure to solve any logical and mathematical problem is called an Algorithm.
Example - $[12 \times (1.9 + 2.1)] - 12 = ?^2$

✓ Normal real life Example-

Let us take one simple day-to-day example by writing algorithm for making „Maggi Noodles“ as a food.

Step 1: Start
Step 2: Take pan with water
Step 3: Put pan on the burner
Step 4: Switch on the gas/burner
Step 5: Put maggi and masala
Step 6: Give two minutes to boil
Step 7: Take off the pan
Step 8: Take out the maggi with the help of fork/spoon
Step 9: Put the maggi on the plate and serve it
Step 10: Stop

Further, the way of execution of the program shall be categorized into three ways:

- (i) sequence statements**
- (ii) selection statements**
- (iii) Iteration or looping statements. This is also called as „control structure“.**

(i) Sequence statements

- In this program, all the instructions are executed one after another.

Example 1. Write an algorithm to print „Good Morning“.

Step 1: Start
Step 2: Print „Good Morning“
Step 3: Stop

Example 2. Write an algorithm to find area of a rectangle.

Step 1: Start
Step 2: Take length and breadth and store them as L and B?
Step 3: Multiply by L and B and store it in area
Step 4: Print area
Step 5: Stop

(ii) Selective Statements

- In this program, some portion of the program is executed based upon the conditional test. If the conditional test is true, compiler will

execute some part of the program, otherwise it will execute the other part of the program.

- We have „true“ or „false“ situation in the program

Example 1. Write an algorithm to check whether he is eligible to vote? (more than or equal to 18 years old).

Step 1: Start

Step 2: Take age and store it in age

Step 3: Check age value, if age ≥ 18 then go to step 4 else step 5

Step 4: Print “Eligible to vote” and go to step 6

Step 5: Print “Not eligible to vote”

Step 6: Stop

Example 2. Write an algorithm to check whether given number is +ve, -ve or zero.

Step 1: Start

Step 2: Take any number and store it in n.

Step 3: Check n value, if $n > 0$ then go to step 5 else go to step 4

Step 4: Check n value, if $n < 0$ then go to step 6 else go to step 7

Step 5: Print “Given number is +ve” and go to step 8

Step 6: Print “Given number is -ve” and go to step 8.

Step 7: Print “Given number is zero”

Step 8: Stop.

(iii) Iterative statements

- In some programs, certain set of statements are executed again and again based upon conditional test. i.e. executed more than one time.
- This type of execution is called „looping or iteration“.

Example 1. Write an algorithm to print all natural numbers up to „n“.

Step 1: Start

Step 2: Take any number and store it in n.

Step 3: Store 1 in i

Step 4: Check i value, if $i \leq n$ then go to step 5 else go to step 8

Step 5: Print i

Step 6: Increment i value by 1

Step 7: Go to step 4

Step 8: Stop.

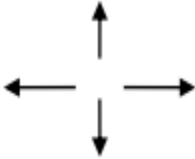
❖ Algorithm characteristics

- 1. Input:** An algorithm may or may not require input
- 2. Output:** Each algorithm is expected to produce at least one result
- 3. Definiteness:** Each instruction must be clear and unambiguous.
- 4. Finiteness:** If the instructions of an algorithm are executed, the algorithm should terminate after finite number of steps
- 5. Independent:** Independent to any programming language

❖ Flowchart

- We know that step-by step process of solving a problem is called algorithm. We can also show these steps in graphical form by using some symbols. This is called flow chart.

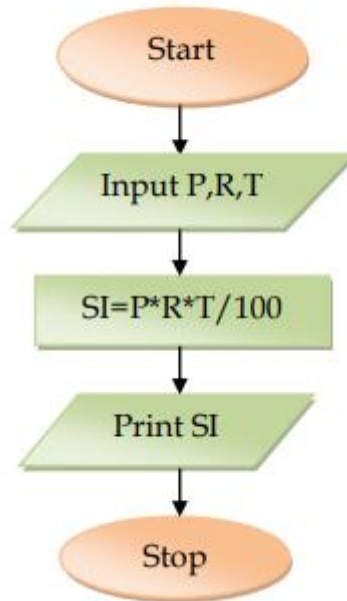
❖ Flowchart Symbols

Symbol Name	Symbol	function
Oval		Used to represent start and end of flowchart
Parallelogram		Used for input and output operation
Rectangle		Processing: Used for arithmetic operations and data-manipulations
Diamond		Decision making. Used to represent the operation in which there are two/three alternatives, true and false etc
Arrows		Flow line Used to indicate the flow of logic by connecting symbols
Circle		Page Connector
		Off Page Connector
		Predefined Process /Function Used to represent a group of statements performing one processing task.
		Preprocessor
		Comments

Example of a sequential execution

1. Draw a flowchart to find the simple interest. (Sequence)

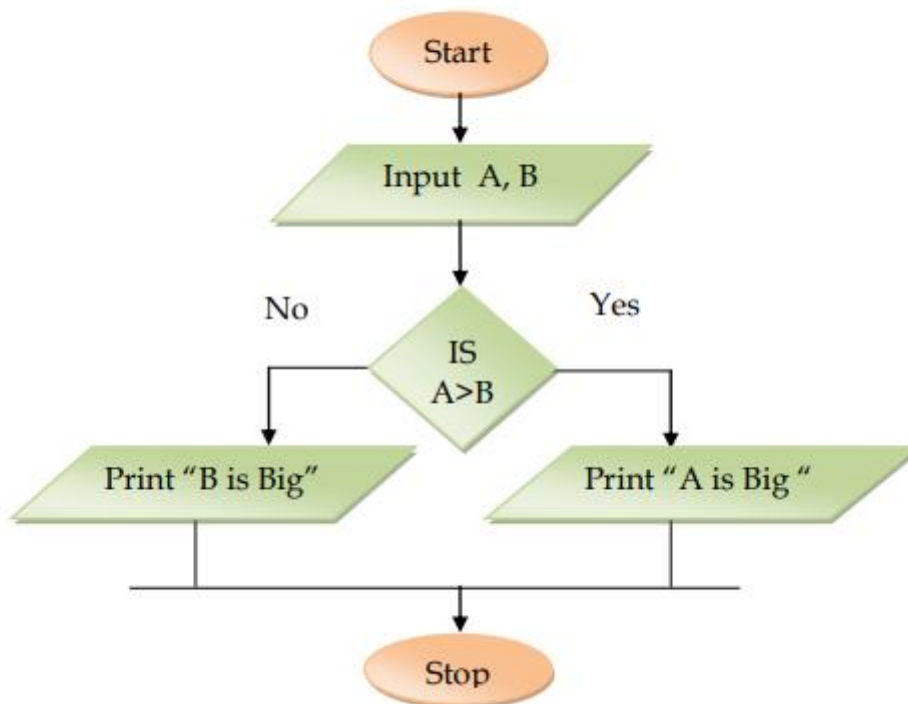
Solution:



Example of a selective execution

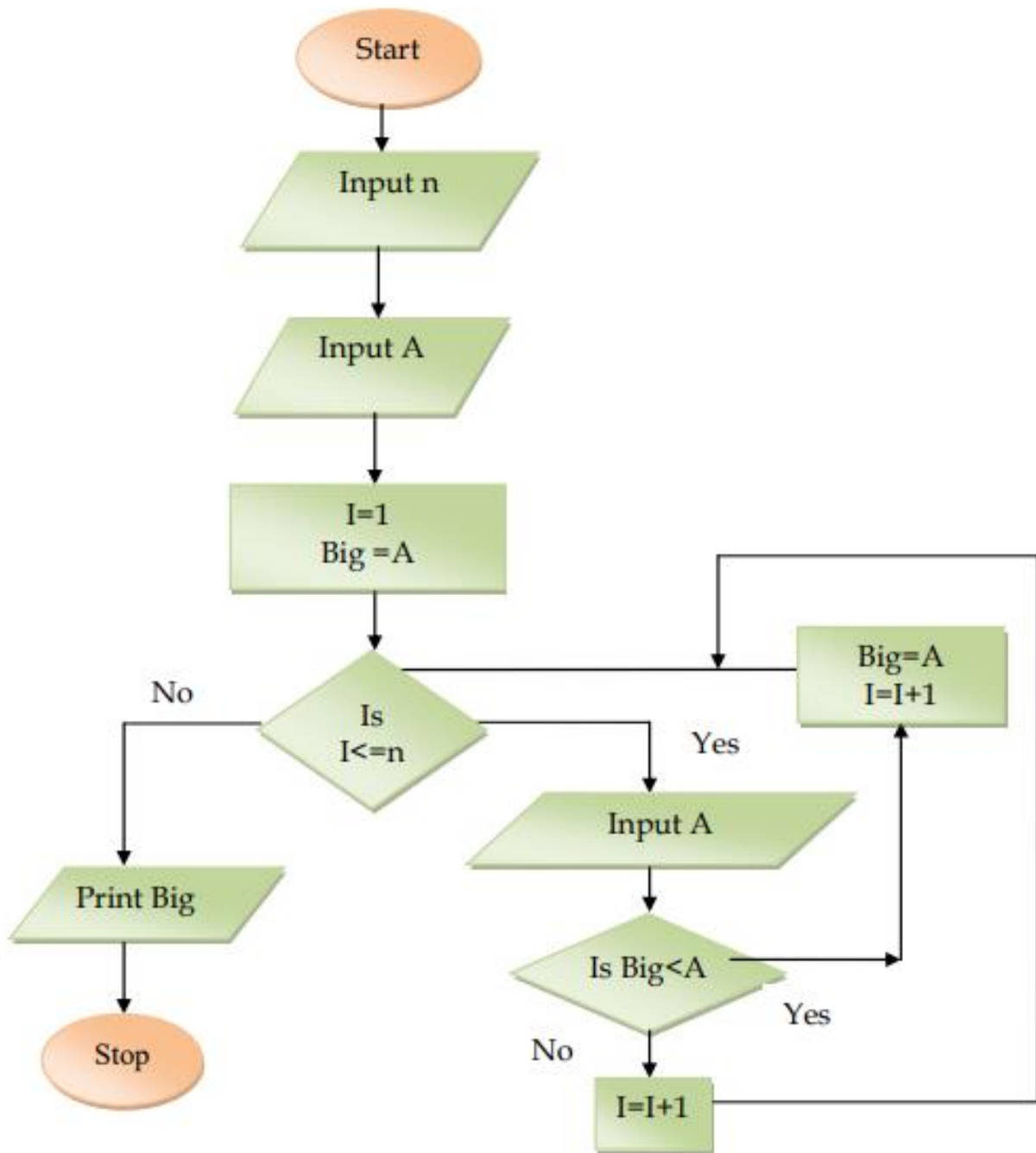
2. Draw a flowchart to find bigger number among two numbers (selective)

Solution:



3. Draw a flow chart to find biggest number among „n“ numbers.

Solution:



Algorithm & Flowchart to find the sum of two numbers

Algorithm

Step-1 Start

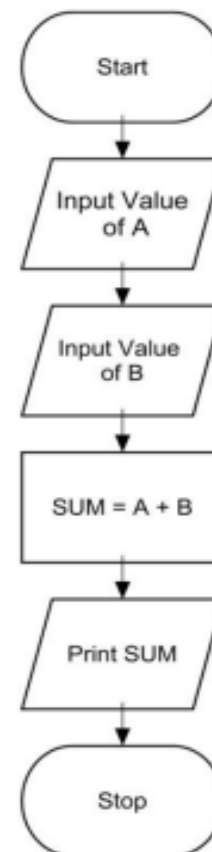
Step-2 Input first numbers say A

Step-3 Input second number say B

Step-4 $SUM = A + B$

Step-5 Display SUM

Step-6 Stop



Algorithm & Flowchart to find Area and Perimeter of Rectangle

L : Length of Rectangle

B : Breadth of Rectangle

AREA : Area of Rectangle

PERIMETER : Perimeter of Rectangle

Algorithm

Step-1 Start

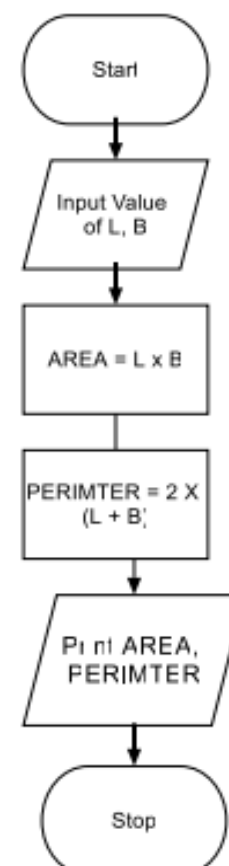
Step-2 Input Side Length & Breadth say L, B

Step-3 $Area = L \times B$

Step-4 $PERIMETER = 2 \times (L + B)$

Step-5 Display AREA, PERIMETER

Step-6 Stop



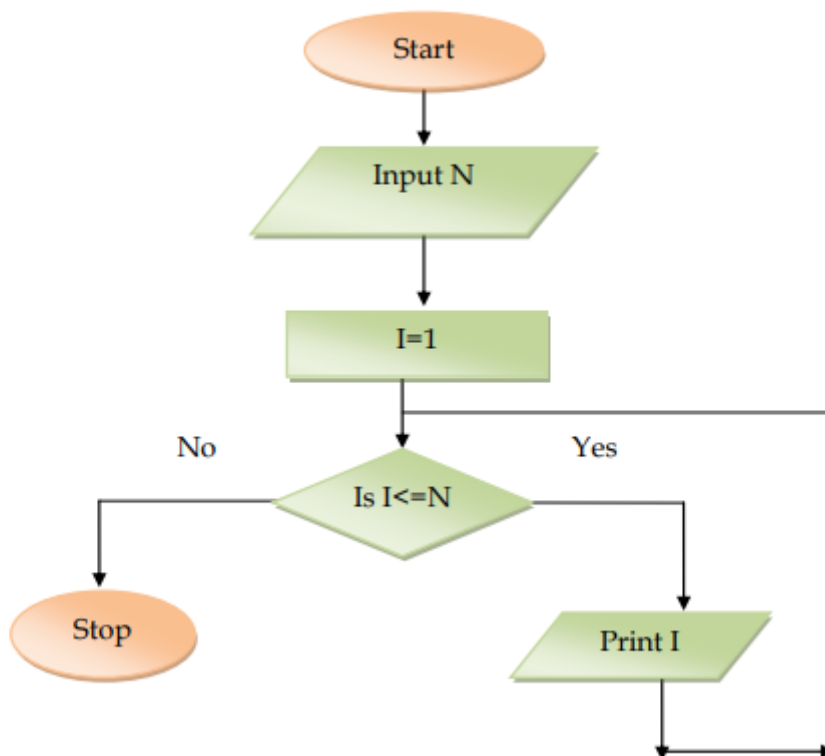
❖ Finite and Infinite loop

- In looping statements, if some set of statements are executed „n“ times (fixed number of times), then it is called „finite loop“.
- At the same time, if some set of statements are executed again and again without any end (infinite times), then it is called „infinite loop“.

Example

Draw a flow chart to print the number from 1 to ∞ .

Solution:



Design a flowchart for the ATM system for withdrawal of cash based on the algorithm defined below

Suppose you went to the ATM to withdraw the amount

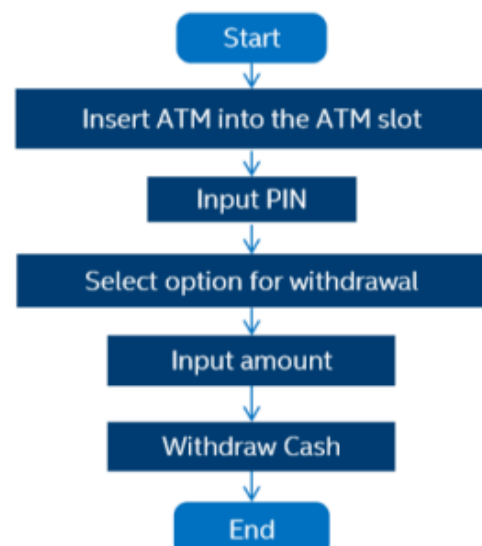
Algorithm:

Step 1 : Insert the ATM card into the ATM slot and remove

Step 2: Input PIN of ATM card

Step 3: Select option of withdrawal

Step 4: Input amount to be withdrawn



Mathematical Operators:

Operator	Meaning	Example
+	Addition	$A + B$
-	Subtraction	$A - B$
*	Multiplication	$A * B$
/	Division	A / B
^	Power	A^3 for A^3
%	Reminder	$A \% B$

Relational Operators

Operator	Meaning	Example
<	Less than	$A < B$
<=	Less than or equal to	$A <= B$
= or ==	Equal to	$A = B$
# or !=	Not equal to	$A \# B$ or $A != B$
>	Greater than	$A > B$
>=	Greater than or equal to	$A >= B$

Logical Operators

Operator	Example	Meaning
AND	$A < B$ AND $B < C$	Result is True if both $A < B$ and $B < C$ are true else false
OR	$A < B$ OR $B < C$	Result is True if either $A < B$ or $B < C$ are true else false
NOT	NOT ($A > B$)	Result is True if $A > B$ is false else true

Selection control Statements

Selection Control	Example	Meaning
IF (Condition) Then ... ENDIF	IF ($X > 10$) THEN $Y = Y + 5$ ENDIF	If condition $X > 10$ is True execute the statement between THEN and ENDIF
IF (Condition) Then ... ELSE ENDIF	IF ($X > 10$) THEN $Y = Y + 5$ ELSE $Y = Y + 8$ $Z = Z + 3$ ENDIF	If condition $X > 10$ is True execute the statement between THEN and ELSE otherwise execute the statements between ELSE and ENDIF

Loop control Statements

Selection Control	Example	Meaning
WHILE (Condition) DO ENDDO	WHILE (X < 10) DO print x x=x+1 ENDDO	Execute the loop as long as the condition is TRUE
DO UNTILL (Condition)	DO print x x=x+1 UNTILL (X >10)	Execute the loop as long as the condition is false

GO TO statement also called unconditional transfer of control statement is used to transfer control of execution to another step/statement. . e.g. the statement GOTO n will transfer control to step/statement n

Algorithm v/s Flowchart

No.	Algorithm	Flowchart
1.	Algorithm is step by step procedure to solve the problem.	Flowchart is a diagram created by different shapes to show the flow of data.
2.	Algorithm is complex to understand.	Flowchart is easy to understand.
3.	In algorithm plain text are used.	In flowchart, symbols/shapes are used.
4.	Algorithm is easy to debug.	Flowchart it is hard to debug.
5.	Algorithm is difficult to construct.	Flowchart is simple to construct.
6.	Algorithm does not follow any rules.	Flowchart follows rules to be constructed.

❖ Pseudocode

- A **Pseudocode** is defined as a step-by-step description of an algorithm. Pseudocode does not use any programming language in its representation instead it uses the simple English language text as it is intended for human understanding rather than machine reading.
- Pseudocode normally uses control statement e.g. While Do, If else, Goto, Endif,.....

Algorithm, Pseudocode & Flowchart - 1

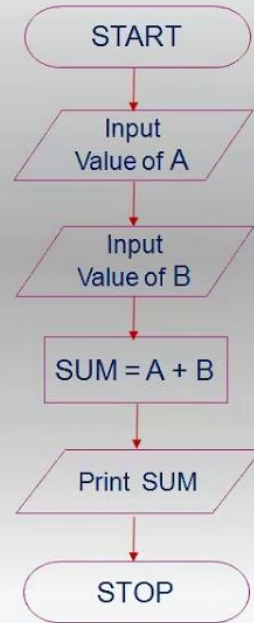
Algorithm & Flowchart to find the sum of two numbers

Algorithm

Step-1 Start
Step-2 Input first numbers say A
Step-3 Input second number say B
Step-4 $SUM = A + B$
Step-5 Display SUM
Step-6 Stop

Pseudocode

Begin
Input A
Input B
Compute $SUM = A + B$
Print SUM
End



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